

COMPREHENSIVE SEPSIS RISK ASSESSMENT

Sentinel AI Clinical Decision Support System

Patient: John Doe

Report Date: June 23, 2026 at 15:27 UTC

Facility: Sentinel AI Medical Center

Patient ID: PID-000000

Ordering Provider: Clinical AI System

Department: Critical Care / Emergency

1. EXECUTIVE SUMMARY & RISK STRATIFICATION

This section provides the top-level sepsis risk probability calculated by the Sentinel AI predictive model based on the patient's current physiologic and laboratory parameters.

14.0%

Risk Category: STABLE

2. AI-GENERATED CLINICAL NARRATIVE

The following narrative translates the quantitative feature attributions into a qualitative clinical assessment, highlighting the primary physiologic drivers of the current risk score.

Current sepsis probability is 14.0%, within stable parameters. Risk is driven primarily by high Heart Rate (110.0 bpm), low MAP (62.0 mmHg) and high Glucose (168.0 mg/dL).

3. COMPREHENSIVE PHYSIOLOGIC PROFILE

A complete breakdown of the analyzed physiological and laboratory parameters, comparing the patient's current values against standard clinical reference ranges.

Parameter	Measured Value	Reference Range	Clinical Status
Glucose	168.0 mg/dL	70.0–140.0 mg/dL	▲ ELEVATED
Heart Rate	110.0 bpm	60.0–100.0 bpm	▲ ELEVATED
MAP	62.0 mmHg	70.0–100.0 mmHg	▼ REDUCED
Oxygen Saturation	89.0 %	95.0–100.0 %	▼ REDUCED
Respiratory Rate	26.0 breaths/min	12.0–20.0 breaths/min	▲ ELEVATED
Temperature	39.0 °C	36.4–37.6 °C	▲ ELEVATED
WBC	14.6 K/ μ L	4.0–11.0 K/ μ L	▲ ELEVATED

4. EXPLAINABLE AI: FEATURE ATTRIBUTION

This section utilizes SHAP (SHapley Additive exPlanations) values to quantify the impact of each variable on the final risk prediction. Higher scores indicate a stronger contribution to the model's output.

Risk Factor	Value	Status	Relative Impact (SHAP)	Score
Heart Rate	110.0 bpm	▲ HIGH		0.500
MAP	62.0 mmHg	▼ LOW		0.467
Glucose	168.0 mg/dL	▲ HIGH		0.200
Temperature	39.0 °C	▲ HIGH		0.000
Oxygen Saturation	89.0 %	▼ LOW		0.000
Respiratory Rate	26.0 breaths/min	▲ HIGH		0.000
WBC	14.6 K/ μ L	▲ HIGH		0.000

5. CLINICAL ACTION PLAN & PROTOCOL PATHWAYS

Based on the identified physiological derangements, the following evidence-based interventions are recommended in alignment with the Surviving Sepsis Campaign (SSC) guidelines.

1. Management of abnormal Heart Rate:

Recommendation:

Assess for tachycardia etiology; consider fluid status evaluation and continuous cardiac monitoring.

Protocol Pathway:

Surviving Sepsis Campaign Hour-1 Bundle
2. Management of abnormal MAP:

Recommendation:

Evaluate for IV fluid resuscitation (30 mL/kg crystalloid); consider vasopressor initiation if MAP < 65 mmHg after fluid challenge.

Protocol Pathway:

Surviving Sepsis Campaign Hour-1 Bundle — Hemodynamic Resuscitation
3. Management of abnormal Glucose:

Recommendation:

Initiate insulin therapy protocol targeting glucose 140–180 mg/dL per ICU glycemic management.

Protocol Pathway:

Surviving Sepsis Campaign — Glycemic Control
4. Management of abnormal Temperature:

Recommendation:

Obtain blood cultures before initiating broad-spectrum antibiotics within 1 hour.

Protocol Pathway:

Surviving Sepsis Campaign Hour-1 Bundle
5. Management of abnormal Oxygen Saturation:

Recommendation:

Initiate supplemental oxygen to maintain SpO2 ≥ 94%; assess for respiratory failure.

Protocol Pathway:

Surviving Sepsis Campaign Hour-1 Bundle
6. Management of abnormal Respiratory Rate:

Recommendation:

Assess for respiratory distress; consider arterial blood gas and chest imaging.

Protocol Pathway:

Surviving Sepsis Campaign — Organ Dysfunction Monitoring
7. Management of abnormal WBC:

Recommendation:

Obtain blood cultures and initiate empiric broad-spectrum antibiotics within 1 hour of recognition.

Protocol Pathway:

Surviving Sepsis Campaign Hour-1 Bundle — Antimicrobial Therapy

APPENDIX A: MODEL METHODOLOGY & VALIDATION

A.1 Predictive Algorithm

The Sentinel AI predictive model employs an ensemble machine learning architecture, specifically an optimized Random Forest classifier. It was trained on a comprehensive retrospective cohort of ICU patient records (MIMIC-III dataset), encompassing thousands of critical care encounters. The model continuously evaluates a multivariate feature space including hemodynamics, respiratory parameters, and point-of-care laboratory values.

A.2 Explainability Framework (SHAP)

To ensure clinical transparency, the system utilizes SHapley Additive exPlanations (SHAP). This game-theoretic approach assigns each feature an importance value for a specific prediction. The SHAP score represents the marginal contribution of a specific vital sign to the final risk probability, allowing clinicians to rapidly identify the physiologic derangements driving the alert.

A.3 Guideline Integration

Actionable recommendations are systematically mapped from identified abnormal features to the established pathways of the Surviving Sepsis Campaign (SSC) Hour-1 Bundle. This ensures that the AI output is not only predictive but directly tied to internationally recognized standards of care for sepsis resuscitation and management.

■ **CLINICAL DISCLAIMER:** This report is generated by an AI-based Clinical Decision Support System. It is intended as a supplementary tool to enhance situational awareness and does NOT constitute a medical diagnosis. The model's predictions and recommendations should always be interpreted in the context of the patient's full clinical presentation. All clinical decisions, including the initiation of therapy, must be made by qualified healthcare professionals in accordance with institutional protocols and clinical judgment.

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